

Theory Behind Simulations

Single VO2 Neuristor under Voltage and current sources

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1 Introduction

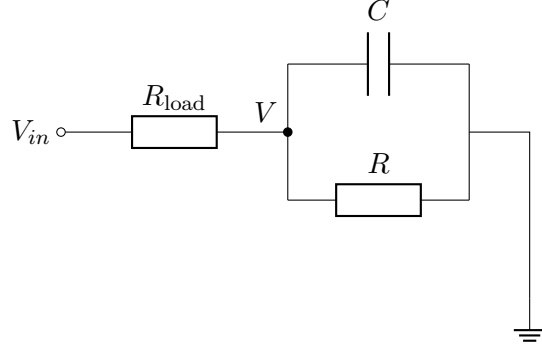


Figure 1: Single-neuristor equivalent circuit used in the voltage-driven model.

1.1 Current Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C \frac{dV_i}{dt} = \frac{V_i^{in}}{R_i^{load}} - V_i \left(\frac{1}{R_i} + \frac{1}{R_i^{load}} \right). \quad (1)$$

Single-device form used in this project:

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}), \quad (2)$$

$$\frac{dV}{dt} = \frac{V_{in} - V}{R_{series}C} - \frac{V}{R_{vo2}C}. \quad (3)$$

Variables and Units

Symbol	Meaning	Units
V_i, V	node/device voltage across VO ₂ and capacitor	V
V_i^{in}, V_{in}	applied source voltage	V
R_i, R_{vo2}	VO ₂ resistance state	Ω
R_i^{load}, R_{series}	series/load resistance	Ω
C	parasitic/effective node capacitance	F
t	time	s
$\mathcal{H}$	hysteresis internal state (branch/reversal memory)	–

Physical Interpretation

- $C dV/dt$: capacitive storage term (how fast node voltage can change).

- Source-branch current is $I_{src} = (V_{in} - V)/R_{series}$, which can be decomposed as $V_{in}/R_{series} - V/R_{series}$.
- V/R_{series} : the part of the source-branch term that opposes charging as node voltage rises.
- V/R_{vo2} : current through the VO₂ device; this term is strongly nonlinear because R_{vo2} depends on temperature and hysteresis state.

The competition between $I_{src} = (V_{in} - V)/R_{series}$ and VO₂ discharge determines oscillatory charging-switching dynamics.

1.2 Thermal Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C_{th} \frac{dT_i}{dt} = \frac{V_i^2}{R_i} - S_e(T_i - T_0) + S_c \nabla^2 T_i + \sigma \eta_i(t). \quad (4)$$

Single-device reduction ($\nabla^2 T = 0$):

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (5)$$

Variables and Units

Symbol	Meaning	Units
T_i, T	device temperature	K
t	time	s
C_{th}	thermal capacitance	J/K
V_i, V	device voltage	V
R_i, R_{vo2}	VO ₂ resistance	Ω
S_e	thermal conductance to environment/substrate	W/K
S_c	nearest-neighbor thermal coupling conductance	W/K
T_0	ambient/base temperature	K
$\nabla^2 T$	discrete thermal Laplacian (neighbor exchange term)	K
$\eta_i(t)$	Gaussian white-noise process (paper notation)	$s^{-1/2}$
σ	noise strength multiplier (paper notation)	$J s^{-1/2}$

Physical Interpretation

- $\frac{V^2}{R_{vo2}}$: Joule-heating input that increases T .
- $S_e(T - T_0)$: heat leakage to ambient that decreases T when $T > T_0$.

- $S_c \nabla^2 T$: diffusive thermal exchange with neighbors; positive when neighbors are hotter.
- $\sigma \eta_i(t)$: stochastic thermal fluctuations (paper noise term).

The thermal equation therefore balances Joule heating, environmental cooling, inter-device heat diffusion, and noise.

1.3 Resistance / Hysteresis Model

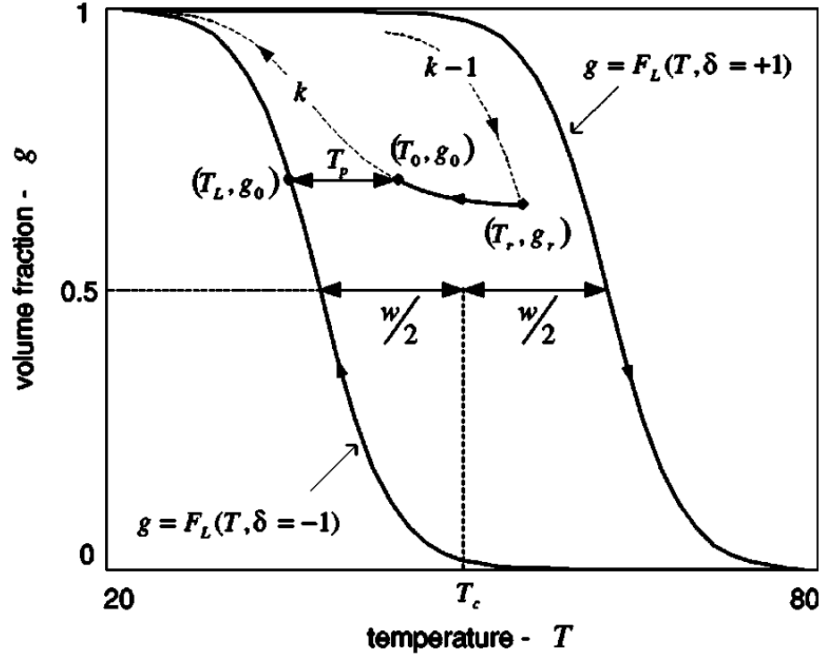


Figure 2: Reference hysteresis geometry showing major branches, reversal points, and proximity temperature shift in the volume-fraction model.

Equation(s)

Starting point from de Almeida *et al.* (2002):

$$R_s(T) = R_0 \exp\left(\frac{E_a}{k_B T}\right). \quad (6)$$

$$R(T) \approx g(T)R_s(T) + (1 - g(T))R_m \Rightarrow R(T) \approx g(T)R_s(T) + R_m. \quad (7)$$

Major-loop semiconducting fraction:

$$g_{\text{major}}(T, \delta) = \frac{1}{2} + \frac{1}{2} \tanh\left[\beta \left(\delta \frac{w}{2} + T_c - T\right)\right], \quad \delta = \begin{cases} +1, & dT/dt > 0, \\ -1, & dT/dt < 0. \end{cases} \quad (8)$$

Minor-loop memory terms:

$$T_{pr} = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_r - 1) - T_r, \quad (9)$$

$$x = \frac{T - T_r}{T_{pr}}, \quad P(x) = \frac{1}{2} (1 - \sin(\gamma x)) (1 + \tanh(\pi^2 - 2\pi x)), \quad T_p = T_{pr} P(x), \quad (10)$$

$$g(T, \mathcal{H}) = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - (T + T_p) \right) \right]. \quad (11)$$

Final resistance law used in this project:

$$R_{vo2}(T, \mathcal{H}) = R_0 \exp \left(\frac{E_a/k_B}{T} \right) g(T, \mathcal{H}) + R_m. \quad (12)$$

Variables and Units

Symbol	Meaning	Units
$R_{vo2}, R(T)$	VO ₂ device resistance	Ω
$R_s(T)$	semiconducting-branch resistance (Arrhenius)	Ω
R_m	metallic-state resistance floor	Ω
R_0	Arrhenius prefactor	Ω
E_a/k_B	activation-energy-over-Boltzmann term	K
T	device temperature (our model: Kelvin)	K
$g(T, \mathcal{H})$	semiconducting volume fraction with hysteresis memory	–
w	hysteresis width parameter	K
T_c	hysteresis center temperature	K
β	branch steepness	K ^{−1}
δ	branch sign (+1 heating, −1 cooling)	–
T_r, g_r	last reversal state (temperature, fraction)	K, –
T_{pr}	reversal-derived proximity temperature scale	K
γ	proximity-function shape parameter	–
$P(x)$	proximity function controlling minor-loop approach	–
T_p	instantaneous proximity temperature shift	K
$\mathcal{H}$	full hysteresis state ($\delta, T_r, g_r, T_{pr}, \dots$)	–
$\tilde{R}, \tilde{g}$	measured resistance and inferred fraction for fitting	$\Omega, -$

Physical Interpretation

- Eq. (6): $R_s(T)$ decreases exponentially with temperature in the insulating/semiconducting regime.
- Eq. (7): total resistance combines semiconducting contribution gR_s with a metallic floor R_m .
- Eq. (8): two major branches are selected by δ , creating the loop width w .

- Eqs. (9)–(10): reversal variables (T_r, g_r, T_{pr}) and $P(x)$ encode memory and generate minor loops.
- Eq. (12): this is the resistance passed to the electrical/thermal ODEs at every timestep.

Assumptions

- $R_s \gg R_m$ in the transition regime justifies the additive form in Eq. (7).
- Major-loop branches are represented by a symmetric tanh form around T_c .
- Hysteresis state updates only at reversal events (dT/dt sign change).

2 Derivation of the Current-Source Model

2.1 Starting Point: Original Voltage-Driven Equations

Equation(s)

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}). \quad (13)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma\eta(t). \quad (14)$$

2.2 Assumptions and Circuit Interpretation

Assumptions

- The voltage source is removed and replaced by a current source in a separate experiment.
- The source enforces an ideal programmed current waveform $I_{in}(t)$.
- An external resistor R_{series} may be present in the injection path in the physical setup.
- Under ideal current drive, adding/removing this external R_{series} does not change node-current balance, so both circuits are dynamically equivalent.
- The VO₂ branch representation is unchanged: $R_{vo2}(T, \mathcal{H})$ in parallel with C , referenced to ground.
- Thermal dynamics keeps the same Joule-heating/cooling structure as in Section 1.2.

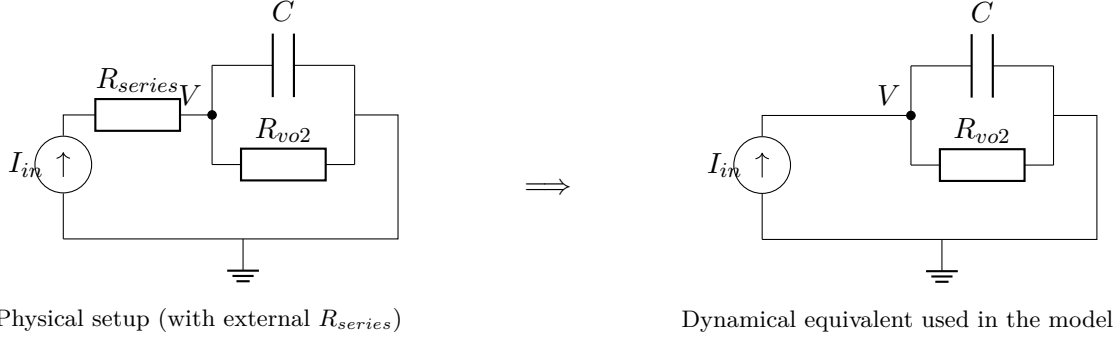


Figure 3: Current-driven setup used in practice (left) and the reduced equivalent used for derivation (right). For ideal enforced $I_{in}(t)$, the external series resistor does not alter node dynamics.

Physical Interpretation

- The imposed source current $I_{in}(t)$ sets the net injected current into node V .
- An external series resistor may drop voltage in the real wiring, but it does not change injected current when the source is ideal.
- At node V , current splits between the capacitor branch and the VO_2 branch.
- VO_2 current V/R_{vo2} remains the electrical-to-thermal coupling path through V^2/R_{vo2} .

2.3 Step-by-Step Derivation for the Direct Current-Source Experiment

Derivation Steps

$$\text{KCL at node } V : \quad I_{in}(t) = I_C + I_{vo2} = C \frac{dV}{dt} + \frac{V}{R_{vo2}}. \quad (15)$$

$$\Rightarrow \quad C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}. \quad (16)$$

This is identical whether or not an external series resistor is included in the physical current-source lead, provided the source enforces $I_{in}(t)$.

2.4 Final Continuous-Time Current-Driven Model

Equation(s)

$$C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}, \quad R_{vo2} = R(T, \mathcal{H}). \quad (17)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (18)$$

Variables and Units

Symbol	Meaning	Units
V	node voltage across VO ₂ and capacitor	V
$I_{in}(t)$	injected source current waveform	A
R_{series}	optional external series resistor in the physical setup (non-dynamical here)	Ω
R_{vo2}	hysteretic VO ₂ resistance $R(T, \mathcal{H})$	Ω
C	electrical capacitance at the node	F
T	device temperature	K
C_{th}	thermal capacitance	J/K
S_e	thermal conductance to environment	W/K
T_0	ambient/base temperature	K
σ	thermal-noise amplitude (current module convention)	W $\sqrt{s}$
$\eta(t)$	white-noise term	–
$\mathcal{H}$	hysteresis internal state	–

3 Discretization

3.1 Discretization of Current Dynamics Equation

Discrete Update Equation

Using forward Euler with time step Δt , and $R_n = R(T_n, \mathcal{H}_n)$:

$$\left. \frac{dV}{dt} \right|_n = \frac{V_{in,n} - V_n}{R_{series}C} - \frac{V_n}{R_nC}. \quad (19)$$

$$V_{n+1} = V_n + \Delta t \left. \frac{dV}{dt} \right|_n. \quad (20)$$

Equivalent compact form:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left[\frac{V_{in,n}}{R_{series}} - V_n \left(\frac{1}{R_n} + \frac{1}{R_{series}} \right) \right]. \quad (21)$$

State Update Order

1. Evaluate hysteretic resistance R_n from current thermal state T_n and hysteresis memory $\mathcal{H}_n$.
2. Compute dV/dt at step n from Eq. (19).
3. Update node voltage to V_{n+1} (explicit Euler).

3.2 Discretization of Thermal Dynamics Equation

Discrete Update Equation

Define $P_n = V_n^2/R_n$. The thermal derivative at step n is:

$$\left. \frac{dT}{dt} \right|_n = \frac{P_n - S_{env}(T_n - T_0) + S_{couple} \nabla_d^2 T_n}{C_{th}} + \xi_n. \quad (22)$$

Forward Euler update:

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (23)$$

where ∇_d^2 is the discrete Laplacian and ξ_n is optional additive temperature-rate noise.

For the simplified single-device non-deterministic case ($S_{couple} = 0$):

$$\left. \frac{dT}{dt} \right|_n = \frac{1}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \xi_n, \quad (24)$$

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (25)$$

State Update Order

1. Use electrical state V_n and resistance R_n to compute Joule power $P_n = V_n^2/R_n$.
2. Compute thermal derivative from Eq. (22) (or Eq. (24) in the reduced case).
3. Update temperature to T_{n+1} (explicit Euler).

3.3 Discretization of Resistance / Hysteresis Model

Discrete State Definitions

$$\mathcal{H}_n = \left\{ \delta_n, T_{r,n}, g_{r,n}, T_{pr,n}, \text{reversed}_n, T_n^{last} \right\}. \quad (26)$$

Given T_n and $\mathcal{H}_n$, the discrete evaluation is:

$$T_{p,n} = T_{pr,n} P \left(\frac{T_n - T_{r,n}}{T_{pr,n}} \right) \text{reversed}_n, \quad (27)$$

In implementation, a small ε regularization is used in the denominator when $|T_{pr,n}|$ is very small.

$$g_n = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta_n \frac{w}{2} + T_c - (T_n + T_{p,n}) \right) \right], \quad (28)$$

$$R_n = R_0 \exp \left(\frac{E_a/k_B}{T_n} \right) g_n + R_m. \quad (29)$$

Update Order (including reversals)

1. Compute $\Delta T_n = T_n - T_n^{last}$. If $|\Delta T_n|$ exceeds threshold and $\text{sign}(\Delta T_n) \neq \delta_n$, trigger reversal.

2. At reversal, set:

$$\delta_n \leftarrow \text{sign}(\Delta T_n), \quad g_{r,n} \leftarrow g(T_n, \mathcal{H}_n^-), \quad T_{r,n} \leftarrow T_n, \quad (30)$$

$$T_{pr,n} \leftarrow \delta_n \frac{w}{2} + T_c - \frac{1}{\beta} \text{arctanh}(2g_{r,n} - 1) - T_{r,n}, \quad \text{reversed}_n \leftarrow 1. \quad (31)$$

3. Evaluate g_n and R_n from Eqs. (28)–(29).

4. Set $T_{n+1}^{last} \leftarrow T_n$ for the next reversal check.

3.4 Discretization of the Current-Source Model

Discrete Update Equation

For the direct ideal-current model of Section 2:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left(I_{in,n} - \frac{V_n}{R_n} \right), \quad R_n = R(T_n, \mathcal{H}_n). \quad (32)$$

Thermal update in the current module (Euler-Maruyama in implementation):

$$T_{n+1} = T_n + \frac{\Delta t}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \frac{\sigma}{C_{th}} \sqrt{\Delta t} \zeta_n, \quad \zeta_n \sim \mathcal{N}(0, 1). \quad (33)$$

State Update Order

1. Evaluate R_n from $(T_n, \mathcal{H}_n)$.
2. Update V_{n+1} with Eq. (32).
3. Update T_{n+1} with Eq. (33).
4. Update hysteresis memory using (T_n, T_{n+1}) for the next step.

4 Fitting of the Resistance Model Based on Real Data

4.1 Input Data and Preprocessing

What is directly measured

The raw export provides eight columns per sample:

$$\{T, R, t, I, V_+, V_-, T_+, T_-\}$$

with units $\{\text{K}, \Omega, \text{s}, \text{A}, \text{V}, \text{V}, \text{K}, \text{K}\}$.

Table 1: Representative rows from the experimental dataset (actual values).

T (K)	R (Ω)	t (s)	I (A)	V_+ (V)	V_- (V)	T_+ (K)	T_- (K)
369.8230	17.6701	161.6042	2.0×10^{-6}	5.42×10^{-5}	-1.65×10^{-5}	369.818	369.828
338.0850	30.9270	399.6742	2.0×10^{-6}	6.23×10^{-5}	-6.14×10^{-5}	338.212	337.958
305.7635	3288.1260	642.2995	2.0×10^{-6}	6.55×10^{-3}	-6.61×10^{-3}	305.892	305.635
289.8545	4625.3550	824.9370	2.0×10^{-6}	9.23×10^{-3}	-9.27×10^{-3}	289.845	289.864
305.2965	3485.7080	939.4225	2.0×10^{-6}	6.98×10^{-3}	-6.96×10^{-3}	305.168	305.425
370.1625	17.5121	1424.8130	2.0×10^{-6}	5.63×10^{-5}	-1.38×10^{-5}	370.036	370.289

What is cleaned/derived

Preprocessing is:

1. Remove the units row (the row containing K, Ohm, sec, ...).
2. Convert all columns to numeric and drop invalid rows.
3. Keep rows with valid Temperature and Resistance; sort by Time.
4. If $\text{median}(T) < 200$, interpret T as $^{\circ}\text{C}$ and convert to K.

Then we compute directional masks from

$$\Delta T_n = T_n - T_{n-1}, \quad \text{cooling if } \Delta T_n < 0, \quad \text{heating if } \Delta T_n > 0. \quad (34)$$

For this dataset after cleaning:

$$N = 264, \quad T \in [289.8545, 370.1625] \text{ K}, \quad R \in [17.3765, 4625.355] \Omega.$$

4.2 What We Obtain at Each Step vs What Must Be Assumed

Table 2: Given vs computed vs unknown quantities in the resistance fit pipeline.

Class	Quantities
Given by experiment	$T(t)$, $R(t)$, time stamps, and auxiliary channels (I , V_+ , V_- , T_+ , T_-).
Computed directly from data	ΔT_n , cooling/heating masks, sweep turning point, and dataset bounds ($T_{\min}$, $T_{\max}$, $R_{\min}$, $R_{\max}$).
Unknown after preprocessing (must be fitted or swept)	R_0 , E_a/k_B , R_m , w , T_c , β , start branch (insulator or metal), and optionally γ if enough minor-loop information exists.

4.3 Parameter Seeding and Intermediate Quantities

Step-by-step outputs

The implemented seeding path is:

1. **Seed** ($R_0, E_a/k_B, R_m$) **from extremes.** A robust low-temperature window is selected; then R_m is scanned and $\ln(R - R_m)$ is linearly regressed versus $1/T$. For this dataset:

$$R_0^{seed} = 10.9606, \quad (E_a/k_B)^{seed} = 1753.57 \text{ K}, \quad R_m^{seed} = 0.8688 \text{ } \Omega.$$

2. **Build seed semiconducting fraction.**

$$g_n^{seed} = \text{clip}\left(\frac{R_n - R_m^{seed}}{R_0^{seed} \exp((E_a/k_B)^{seed}/T_n)}, 0, 1\right). \quad (35)$$

3. **Seed major-loop geometry** (w, T_c). Cooling/heating temperatures closest to $g^{seed} = 0.5$ are used:

$$w^{seed} = 5.804 \text{ K}, \quad T_c^{seed} = 331.235 \text{ K}.$$

4. **Seed start branch.** From the first non-zero ΔT : `start_branch_hint = metal`.

4.4 Optimization Objective and Fitting Procedure

Objective function

The optimizer minimizes, for parameter vector x and branch b :

$$\mathcal{L}(x, b) = \text{RMSE}_{\log R}(x, b) + \lambda_g \text{RMSE}_g(x, b) + \mathcal{P}_\gamma, \quad \lambda_g = 0.2. \quad (36)$$

with

$$\text{RMSE}_{\log R} = \sqrt{\frac{1}{N} \sum_{n=1}^N [\log_{10} R_n^{model}(x, b) - \log_{10} R_n^{data}]^2}, \quad (37)$$

$$\text{RMSE}_g = \sqrt{\frac{1}{N} \sum_{n=1}^N [g_n^{model}(x, b) - g_n^{exp}(x)]^2}, \quad (38)$$

$$g_n^{exp}(x) = \text{clip}\left(\frac{R_n^{data} - R_m(x)}{R_0(x) \exp((E_a/k_B)(x)/T_n)}, 0, 1\right). \quad (39)$$

When `--fit-gamma` is active:

$$\mathcal{P}_\gamma = 0.01 |\gamma - \gamma_{\text{fixed}}|,$$

otherwise $\mathcal{P}_\gamma = 0$.

Meaning of Terms and Acronyms

- **RMSE:** *Root Mean Squared Error*. It is the square root of the mean squared residual.
- $\text{RMSE}_{\log R}$: RMSE computed on $\log_{10}(R)$, so multiplicative resistance mismatches are penalized more uniformly across decades.

- RMSE_g : RMSE between model and reconstructed semiconducting fraction g , used to keep hysteresis-shape consistency.
- λ_g : scalar weight that controls the influence of the g -consistency term relative to resistance fitting ($\lambda_g = 0.2$ in this work).
- $\mathcal{P}_\gamma$: optional regularization term for γ , used only when `--fit-gamma` is enabled.
- $\text{clip}(u, 0, 1)$: saturation/projection operator to enforce physical bounds for a fraction:

$$\text{clip}(u, 0, 1) = \begin{cases} 0, & u < 0, \\ u, & 0 \leq u \leq 1, \\ 1, & u > 1. \end{cases}$$

- x : optimization vector containing fitted parameters (e.g., $R_0, E_a/k_B, R_m, w, T_c, \beta$, optionally γ).
- b : initial hysteresis branch choice (`insulator` or `metal`) used when evaluating a candidate x .

This optimizer is a custom implementation. Methodologically, it is a bounded derivative-free search (random exploration + coordinate local refinement), conceptually related to direct-search/pattern-search methods (see Sources 3–5) and standard nonlinear optimization practice (Source 6).

Algorithm

1. **Bounded initialization:** define lower/upper bounds for each component of x , then build initial candidates from (i) baseline values and (ii) data-driven seeds.
2. **Branch screening:** evaluate candidates on both initial branches (`insulator`, `metal`) and keep the currently best pair (x, b) .
3. **Global exploration (random search):** for many iterations, propose new points by either: (a) Gaussian perturbation around current best, or (b) uniform draw in the full box. Every proposal is projected back into bounds componentwise by clipping.
4. **Local refinement (coordinate search):** update one coordinate at a time with $\pm$ step moves; accept only improving moves. If no coordinate improves, shrink step size and continue.
5. **Final branch check:** with the best x , evaluate both branches once more and keep the better one.
6. **Report:** output final parameters and RMSE metrics (overall, cooling-only, heating-only).

4.5 Final Fitted Parameters, Quality Metrics, and Limits

Results

For the fitted case documented here:

Table 3: Fitted resistance/hysteresis parameters.

Parameter	Value
R_0	0.5297358678
E_a/k_B (K)	2658.0234292
R_m (Ω)	18.1978381368
w (K)	6.9003303359
T_c (K)	333.5701800799
β	0.2969914699
γ	0.9562696820 (fixed in this run)
Start branch	metal

Table 4: Fit quality metrics ($\log_{10} R$ RMSE).

Metric	Value
Overall RMSE	0.0375554
Cooling RMSE	0.0343180
Heating RMSE	0.0406742

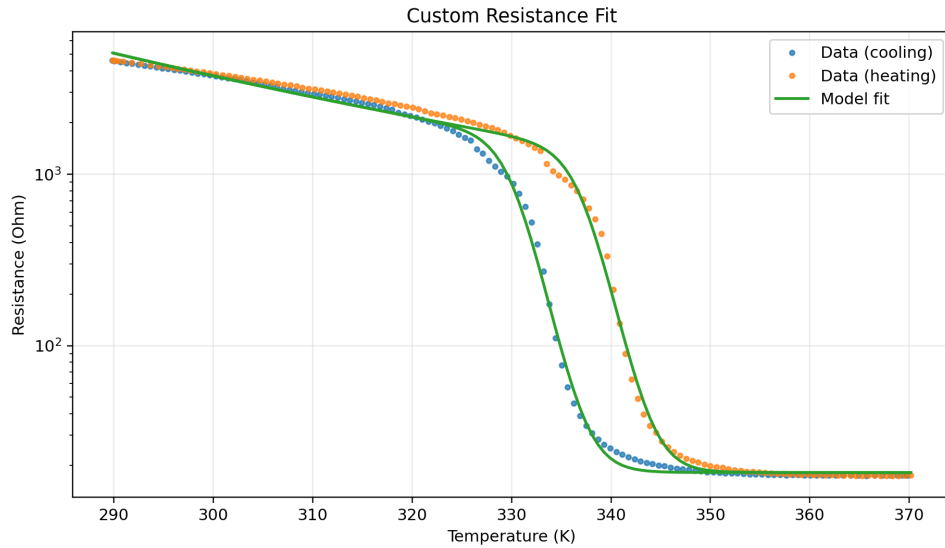


Figure 4: Measured $R(T)$ points and fitted model for the dataset used in Section 4.

Limitations

At the end of this procedure, the following still require additional identification work:

- **Minor-loop fidelity:** one major cooling-heating cycle does not fully constrain history parameters; dedicated reversal experiments are needed.
- **γ identifiability:** with major-loop data only, γ is weakly identifiable, so it is fixed unless additional data is provided.
- **Parameter uniqueness:** different parameter tuples can produce very similar $\log R(T)$ errors; confidence intervals are not yet computed.
- **Not part of this fit:** electrical/thermal dynamical parameters (C , C_{th} , S_e , source terms) must be calibrated separately from transient simulations.

5 Sources

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